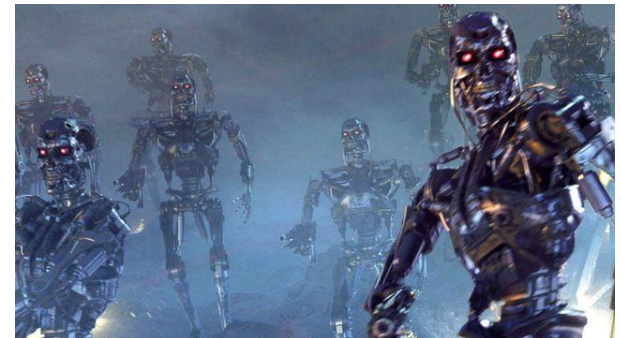
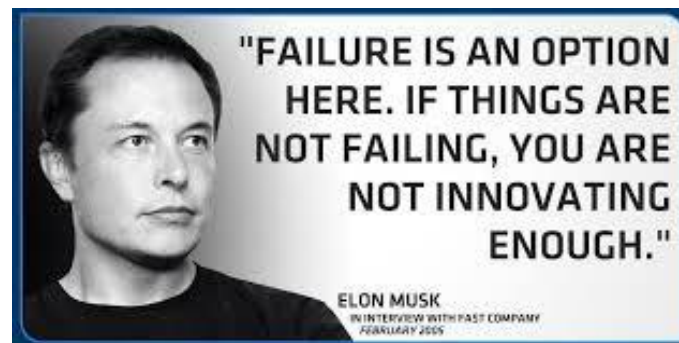




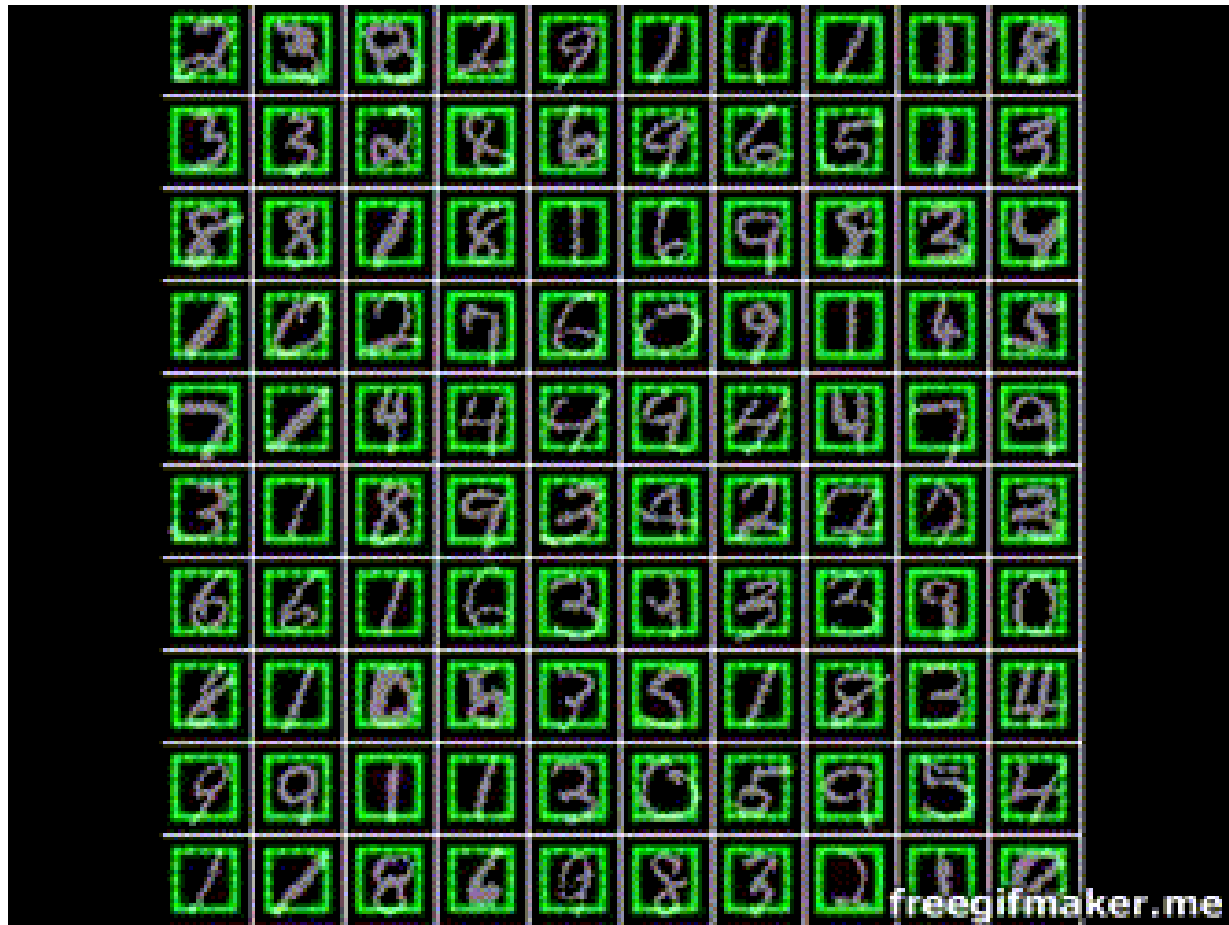
AGI et Finance : l'opportunité des réseaux neuronaux avancés (part2)

28/03/2017 //////////////

Olivier Croissant



DRAW-MNIST



DRAW-2digits



Generating two digits

freegifmaker.me

Vers les AGI



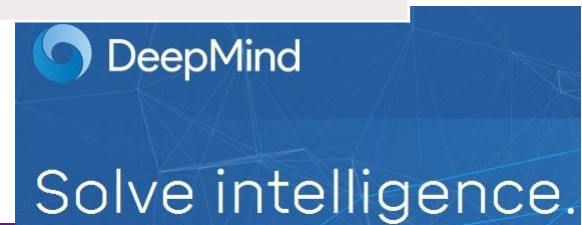
Paradigme	Avantages
Mécanisme d'attention (DRAW)	Simplifier l'inférence par la localisation
LSTM (RNN)	Structurer l'inférence par un raisonnement par étape
Système génératif (VAE)	Généraliser l'inférence grâce aux capacités interpolatoires
Algorithmique différentiable (DNC)	Inférence hétérogène complexe
Mémoires dynamiques (GTMM)	Inférence avec mémoire long terme flexible
Multi-Interfaces neuronaux	Inférence transférable, coopérative et massive

An important distinction

- "Narrow" (classical) AI:
 - Systems explicitly designed to solve specific, reasonably well-defined problems
 - E.g. Deep Blue, Watson, etc.

- Artificial General Intelligence:
 - Systems designed to autonomously learn new tasks and adapt to changing environments

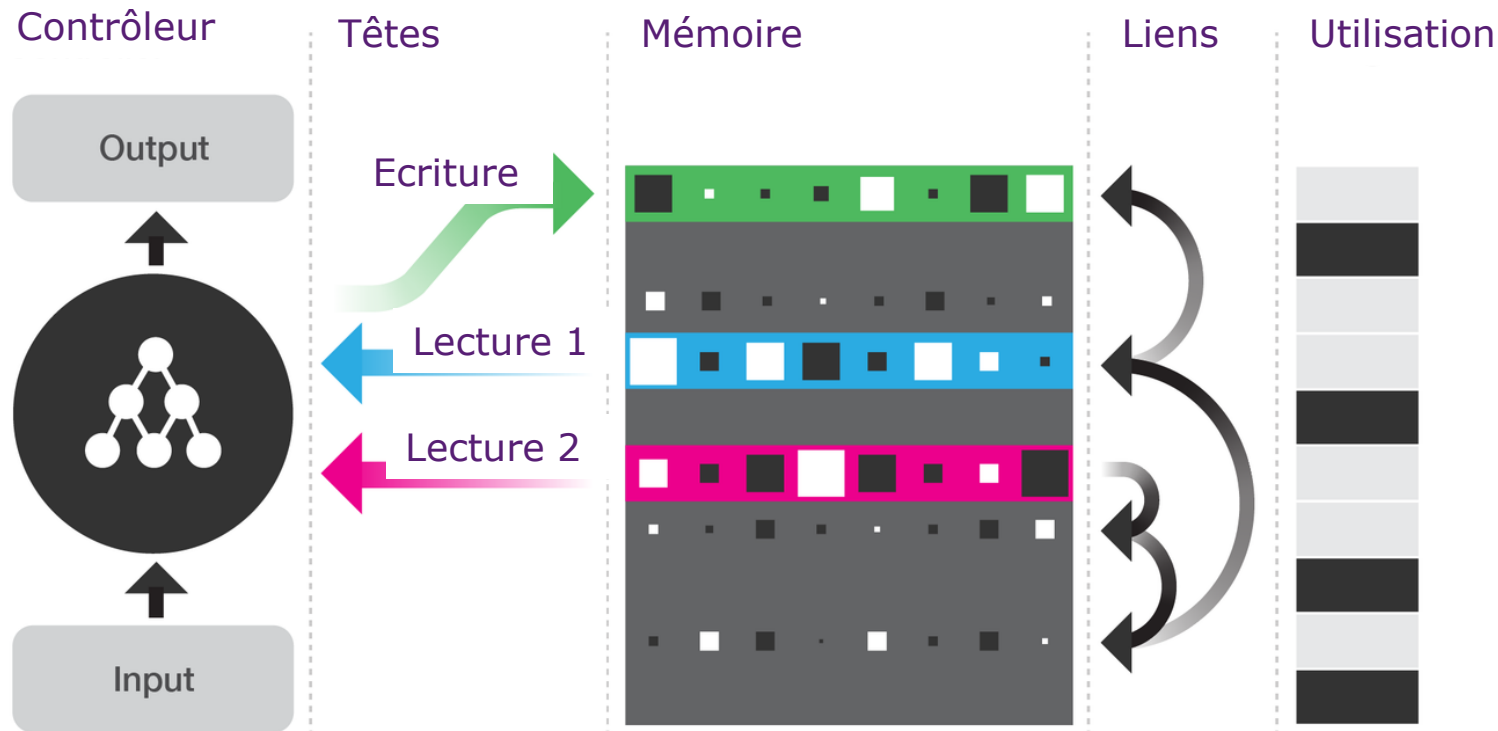
Intellifest 2012



DNC: algorithmique différentiable

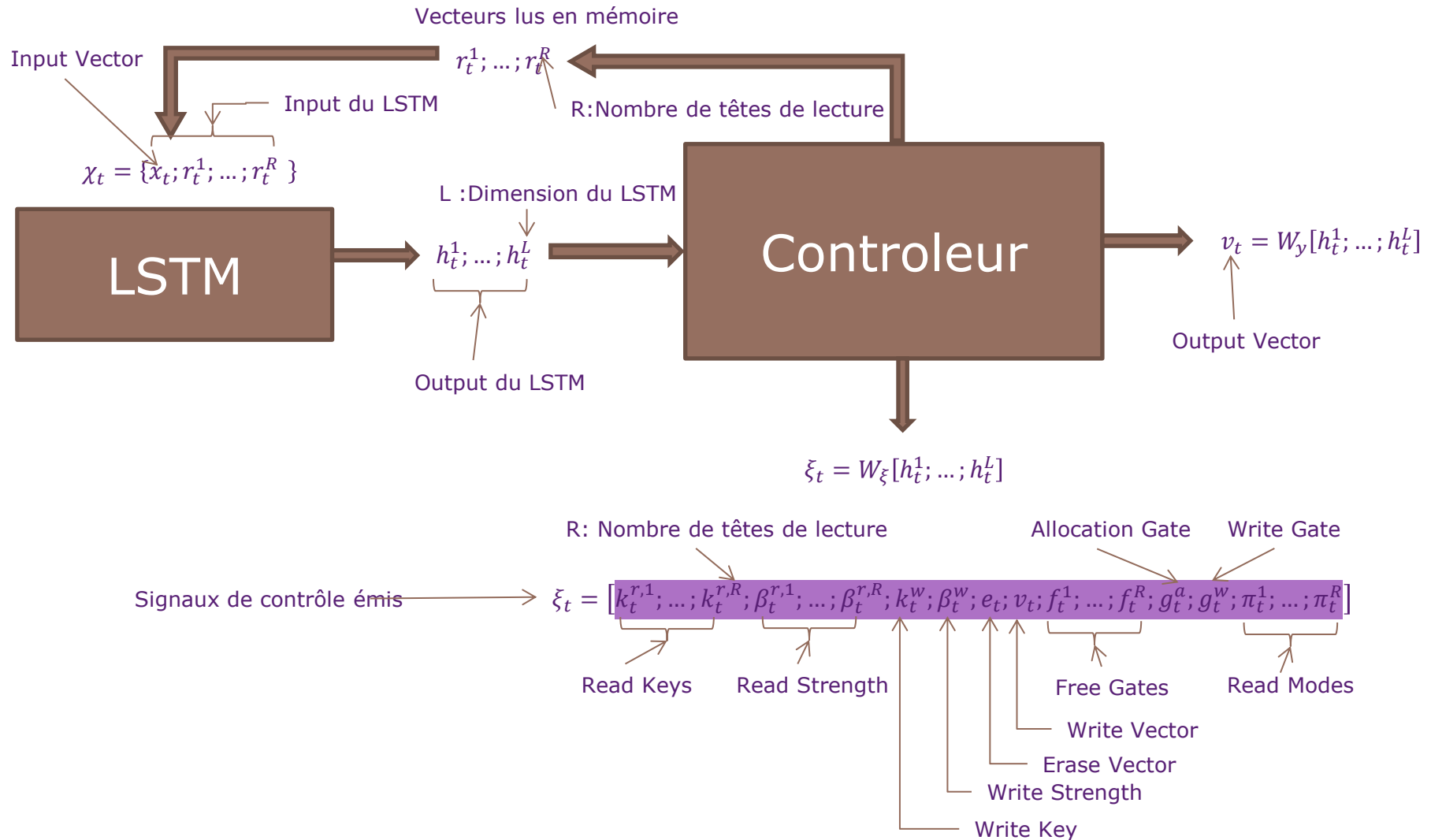


Architecture générale du réseau

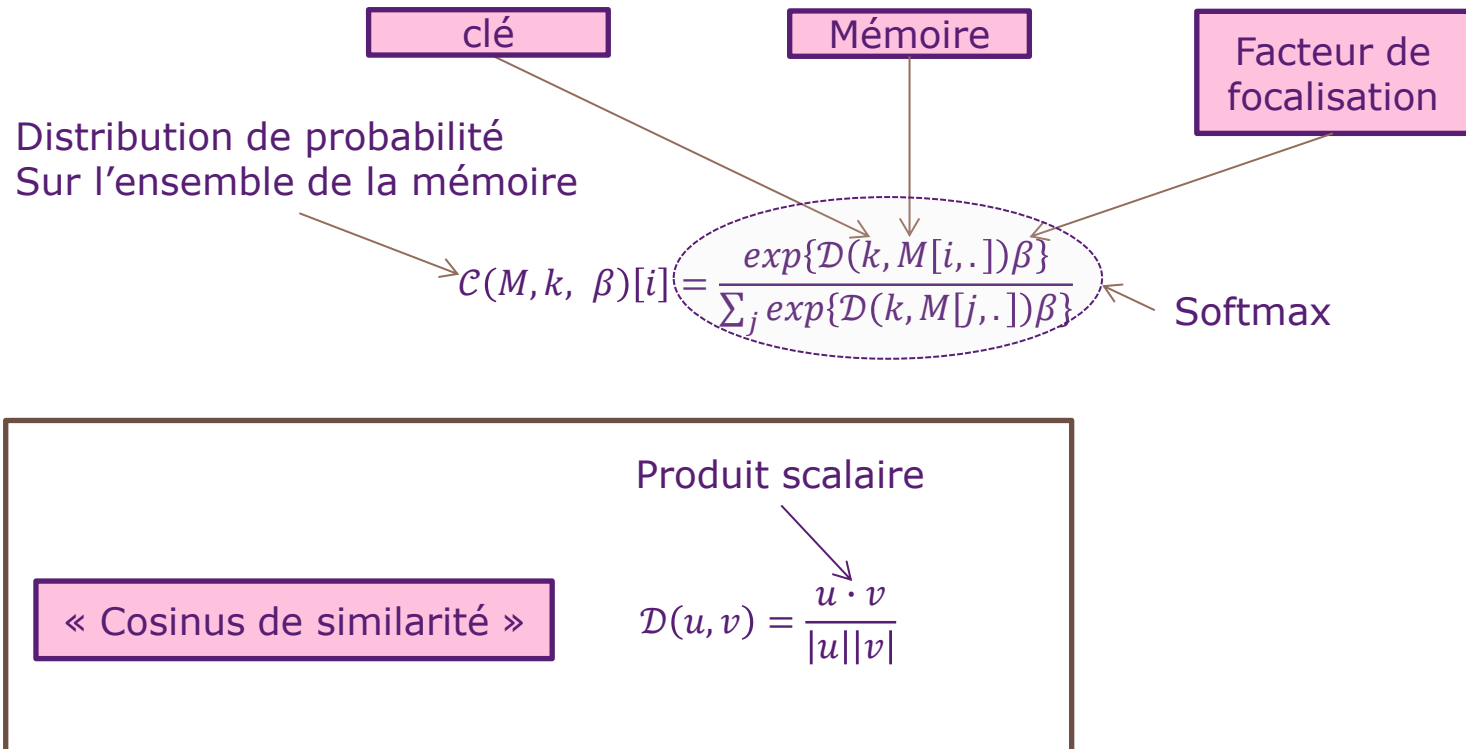


DNC : généralisation du mécanisme d'attention

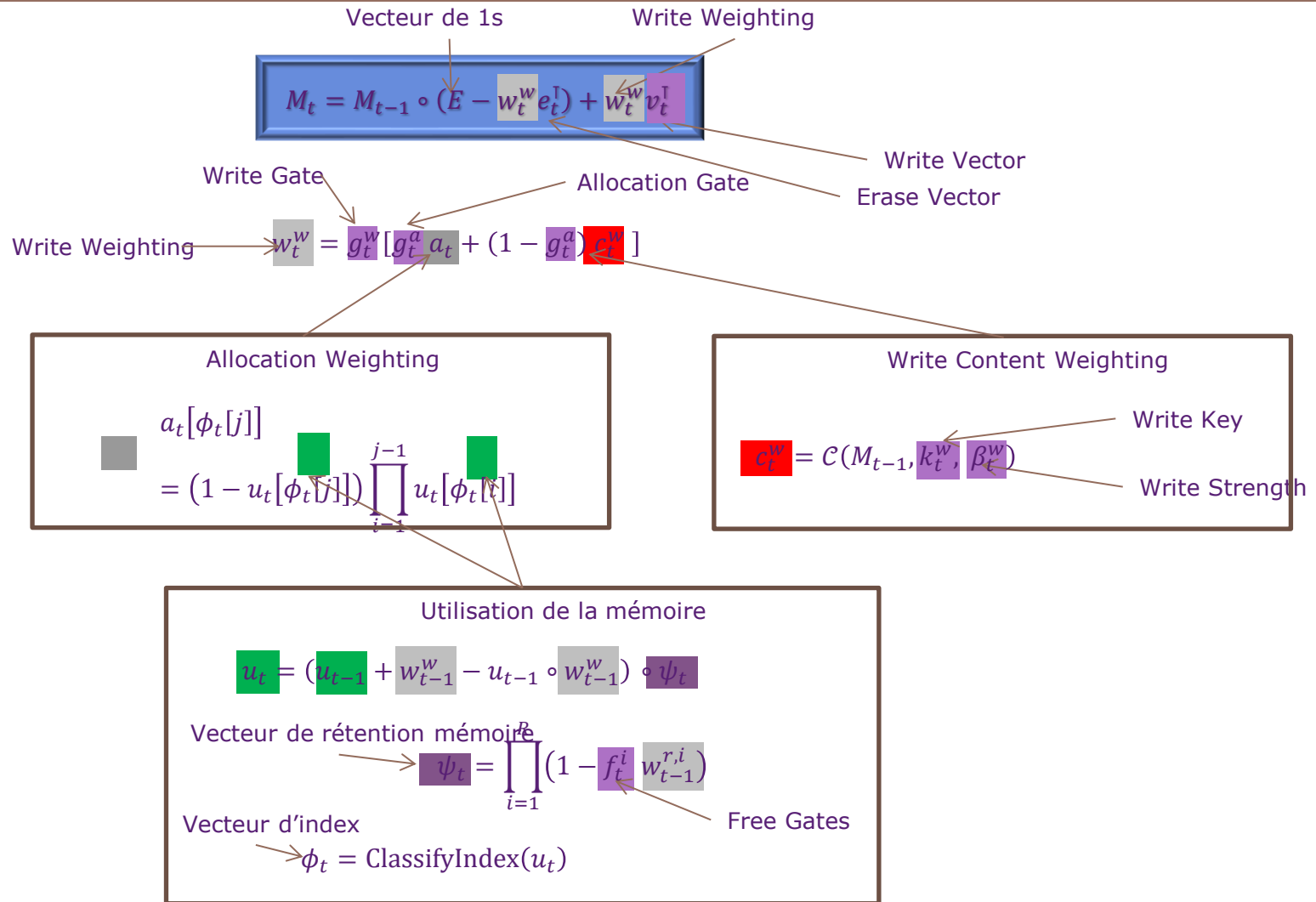
DNC : Contrôleur , Fonctions



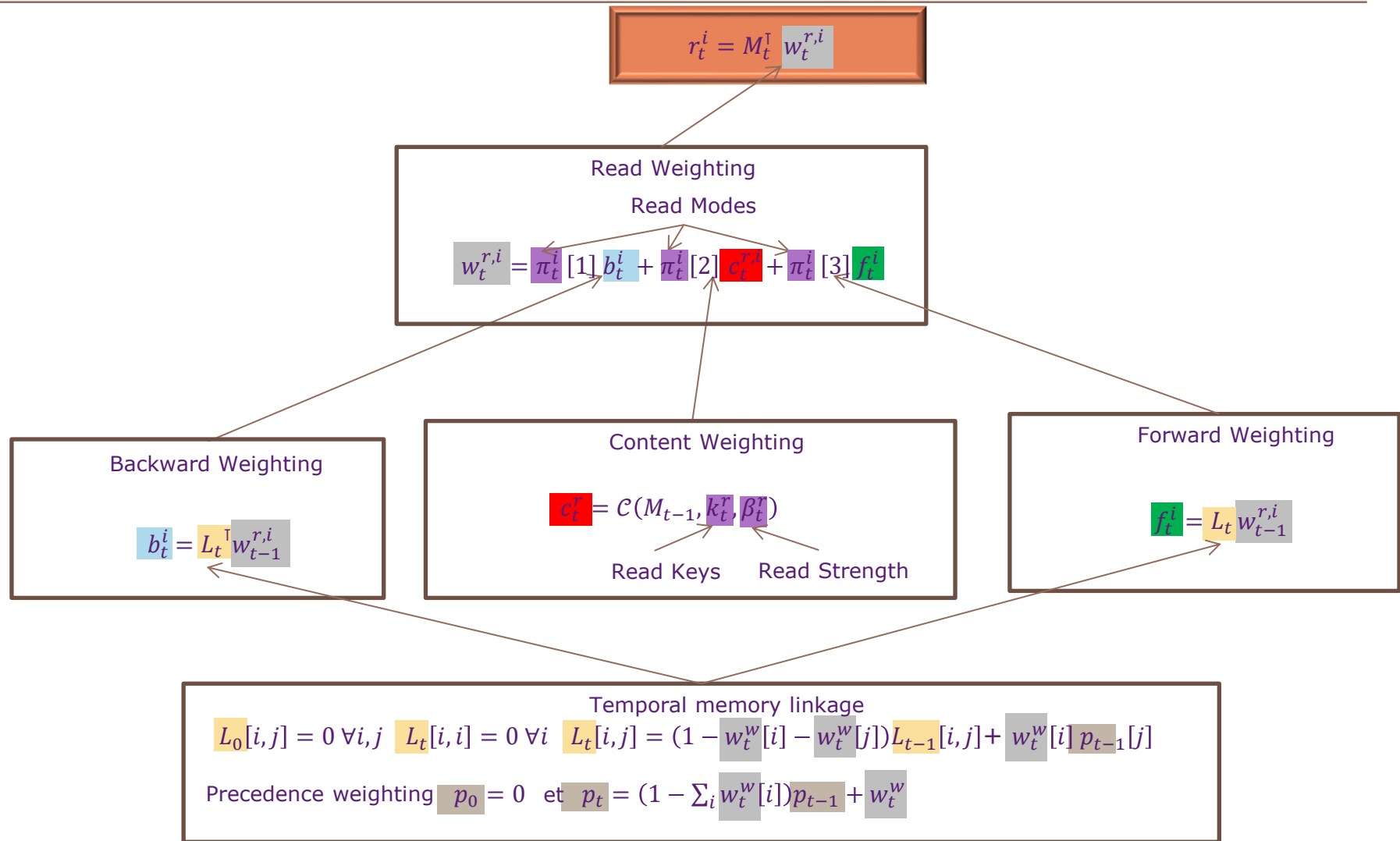
Adressage par contenu de la mémoire



DNC : Contrôleur : Ecriture mémoire



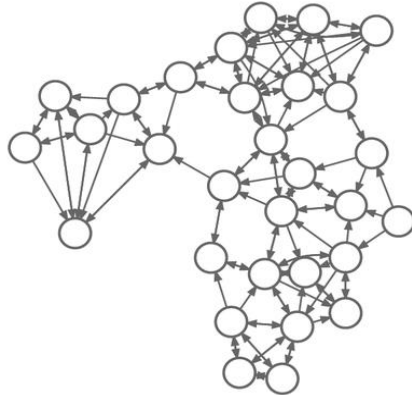
DNC : Contrôleur: lecture mémoire



Problème du voyageur de commerce



Random Training Graph



London Underground



Underground Input:

(OxfordCircus, TottenhamCtRd, Central)
 (TottenhamCtRd, OxfordCircus, Central)
 (BakerSt, Marylebone, Circle)
 (BakerSt, Marylebone, Bakerloo)
 (BakerSt, OxfordCircus, Bakerloo)
 ...
 (LeicesterSq, CharingCross, Northern)
 (TottenhamCtRd, LeicesterSq, Northern)
 (OxfordCircus, PiccadillyCircus, Bakerloo)
 (OxfordCircus, NottingHillGate, Central)
 (OxfordCircus, Euston, Victoria)

- 84 edges in total

Traversal Question:

(BondSt, _, Central),
 (_, _, Circle), (_, _, Circle),
 (_, _, Circle), (_, _, Circle),
 (_, _, Jubilee), (_, _, Jubilee),

Answer:

(BondSt, NottingHillGate, Central)
 (NottingHillGate, GloucesterRd, Circle)
 ...
 (Westminster, GreenPark, Jubilee)
 (GreenPark, BondSt, Jubilee)

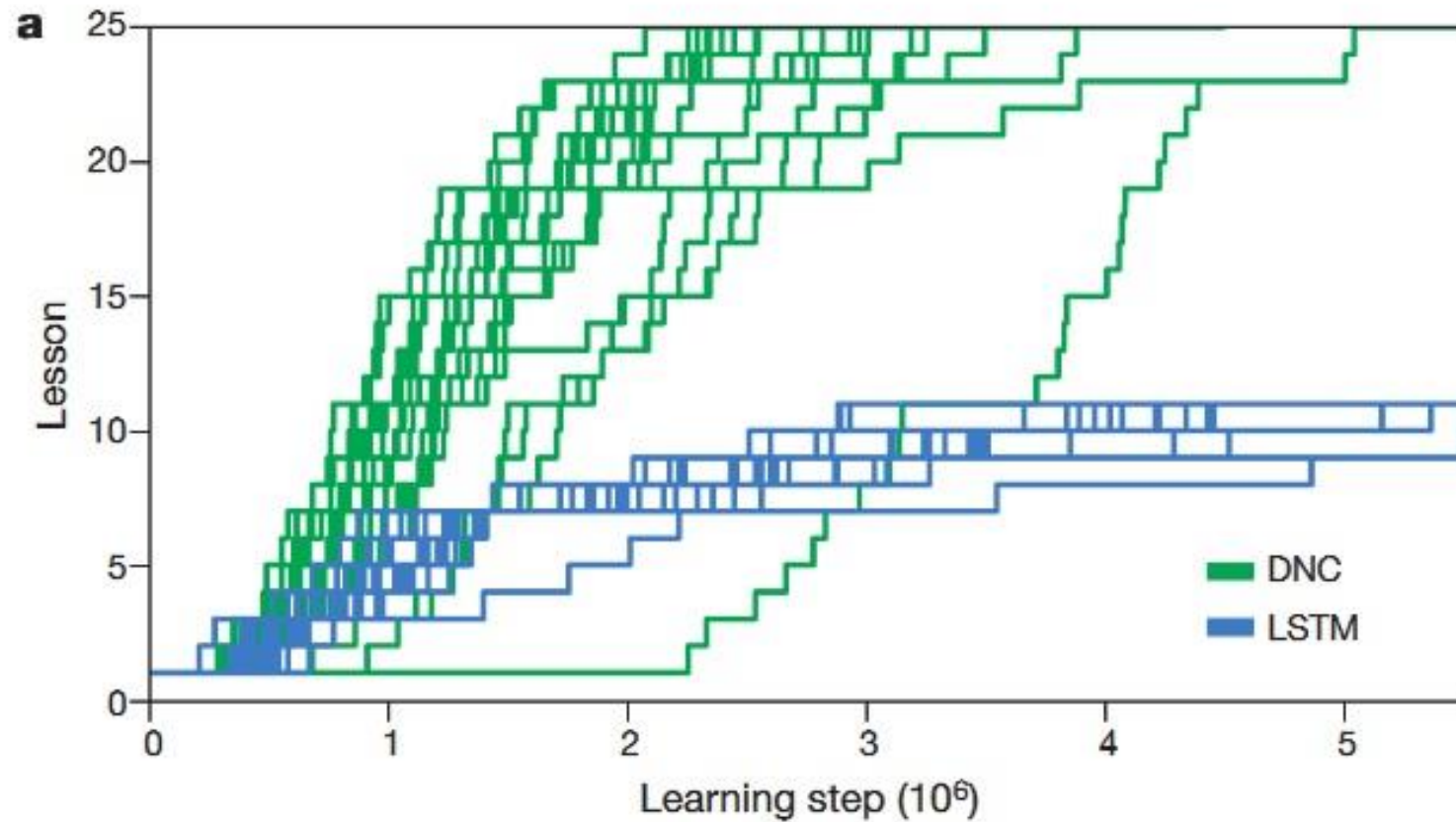
Shortest Path Question:

(Moorgate, PiccadillyCircus, _)

Answer:

(Moorgate, Bank, Northern)
 (Bank, Holborn, Central)
 (Holborn, LeicesterSq, Piccadilly)
 (LeicesterSq, PiccadillyCircus, Piccadilly)

Supériorité du DNC



Question ?



Est-il possible de construire une machine de Turing neurale universelle ?

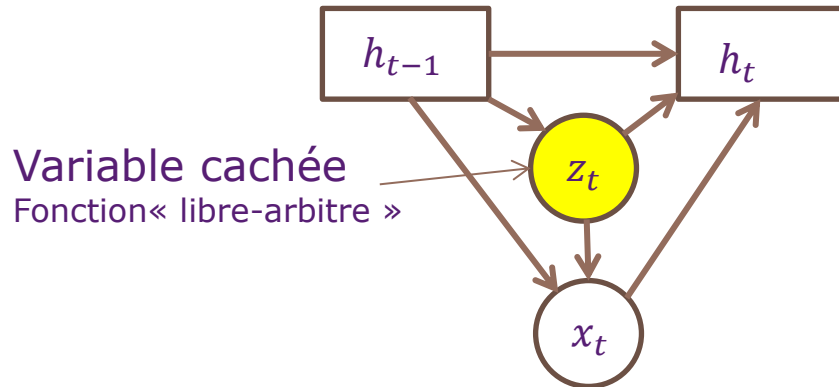
- Elle doit permettre de représenter tout programme
- Elle doit permettre de converger vers tout programme

Comparaison GTMM –Autres



	LSTM	DNC	VRNN	GTMM
Génératif	Non	Non	Oui	Oui
Mémorisation Longue	Oui	Oui	Non	Oui
Scalabilité	Faible	Forte	Faible	Forte
Inférence	Exacte	Exacte	Variationnelle	Variationnelle
Mémoire externe	Non	Oui	Non	Oui
Input Stream de grande dimension	Non	Non	Oui	Oui
Corrélation Capacité mémoire et capacité processing	Forte	Faible	Très forte	Faible

Le variationnel dans les GTM



Reseau	LSTM
Fonction de transition	$h_t = f_h(h_{t-1}, z_t, x_t)$
Fonction prior	$z_t = f_z(h_{t-1})$
Fonction d'observation	$x_t = f_x(h_{t-1}, z_t)$

$$\text{Log } p(x_{\leq T}) = \text{Log} \int p_{\theta}(x_{\leq T}, z_{\leq T}) dz_{\leq T} = \text{Log} \mathbb{E}_{q_{\phi}(z_{\leq T} | x_{\leq T})} \left[\frac{p_{\theta}(x_{\leq T} | z_{\leq T})}{q_{\phi}(z_{\leq T} | x_{\leq T})} \right]$$

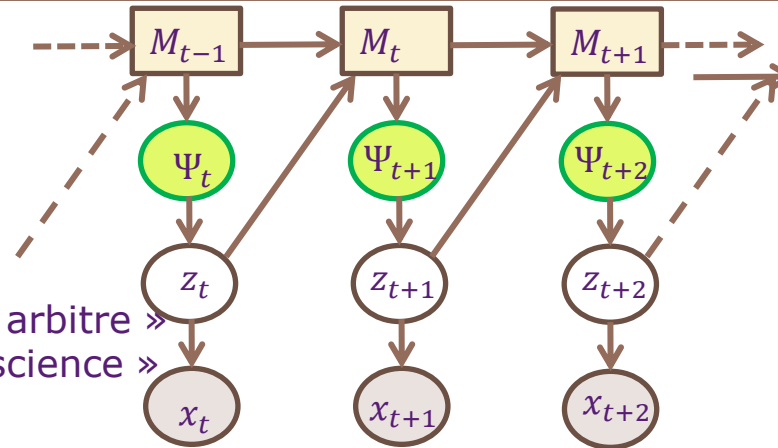
$$\geq \mathbb{E}_{q_{\phi}(z_{\leq T} | x_{\leq T})} [p_{\theta}(x_{\leq T} | z_{\leq T})] - \text{KL}[q_{\phi}(z_{\leq T} | x_{\leq T}) || p_{\theta}(z_{\leq T})] = \mathcal{F}(q; \theta)$$

On choisit un variationnel auto régressif $q_{\phi}(z_{\leq T} | x_{\leq T}) = \prod_{t=1}^T q_{\phi}(z_t | x_{\leq t})$

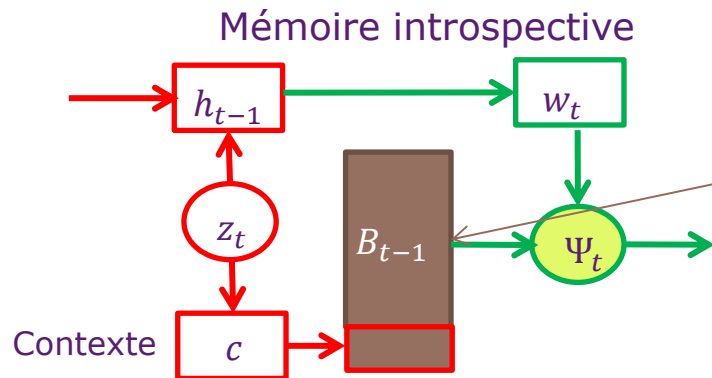
Cela permet d'écrire $\mathcal{F}(q; \theta) = \sum_{t=1}^T \mathcal{F}_t(q; \theta)$

$$\mathcal{F}_t(q; \theta) = \mathbb{E}_{q_{\phi}(z_t | z_{<t}, x_{\leq t})} [p_{\theta}(x_t | z_{\leq t}, x_{<t})] - \text{KL}[q_{\phi}(z_t | z_{<t}, x_{\leq t}) || p_{\theta}(z_t | z_{<t}, x_{<t})]$$

Différents accès mémoriels



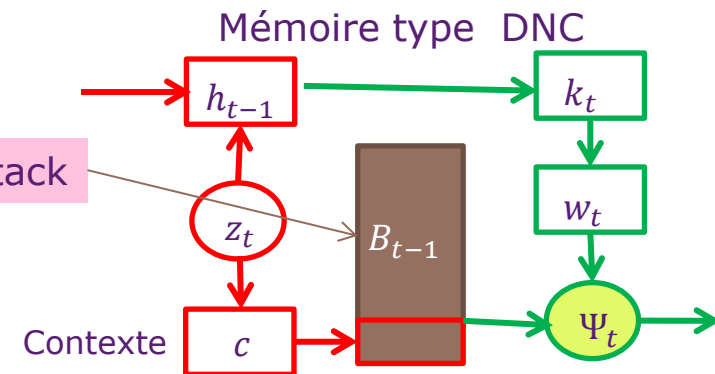
Mémorisation du « libre arbitre »
= « Phénomène de conscience »



Soft attention: $w_t^r = f_{att}(h_{t-1})$

Update de l'état: $h_t = f_{key}(h_{t-1}, z_{t-1}, c_t)$

Avantage : apprentissage rapide



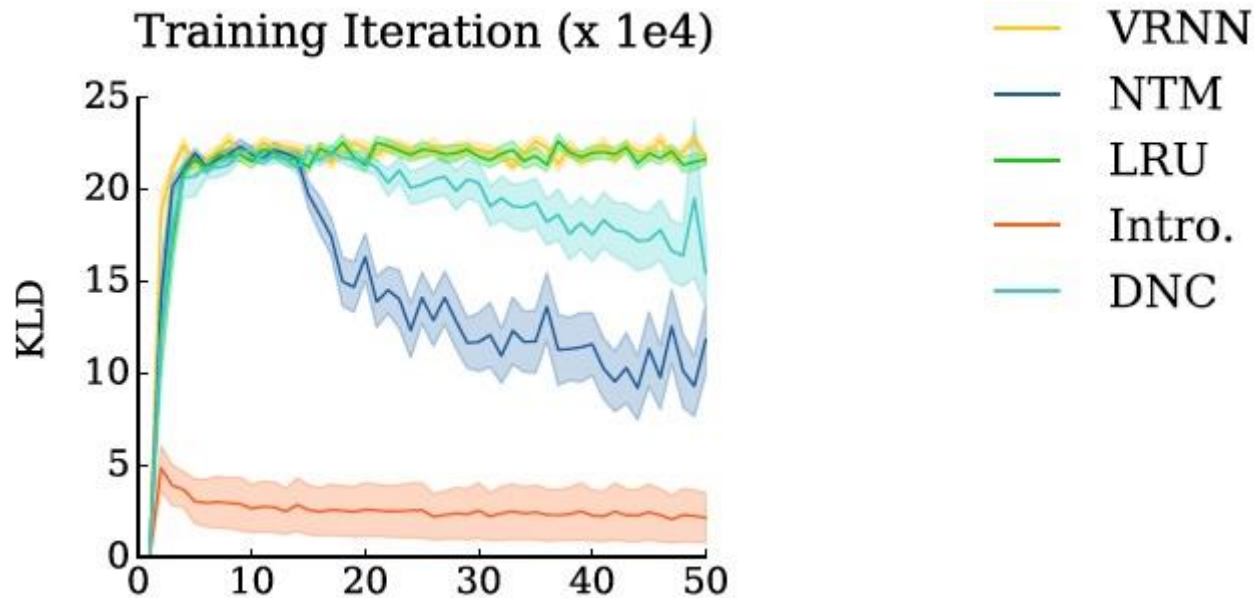
Lecture par contenu: $k_t = f_{key}(h_{t-1})$

$w_t^r = f_{att}(M_{t-1}, k_t)$

Update de l'état: $h_t = f_{key}(h_{t-1}, M_{t-1}, z_{t-1}, c_t)$

Avantage : apprentissage complexe

Supériorité du GTMM avec mémoire introspective

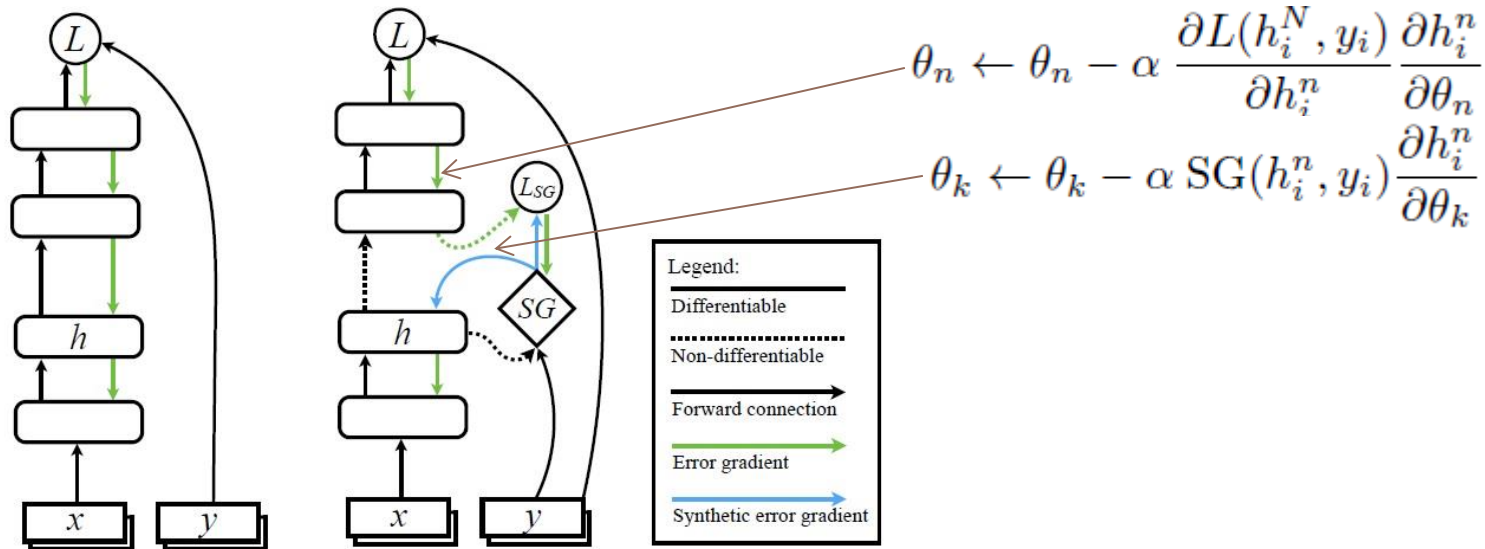


Tache: apprentissage de la répétition d'une séquence de digits

0	1	2	3	4	5	6	7	8	9	PRE-RECALL INTERVAL (l)										RECALL INTERVAL (k)				
3	4	3	8	0	2	4	4	9	4	9	7	0	9	8	1	8	1	0	1	3	4	3	8	0

Stimuli de distraction

Interfaces neuronales Découplés



Backpropagation usuelle Backpropagation avec DNI

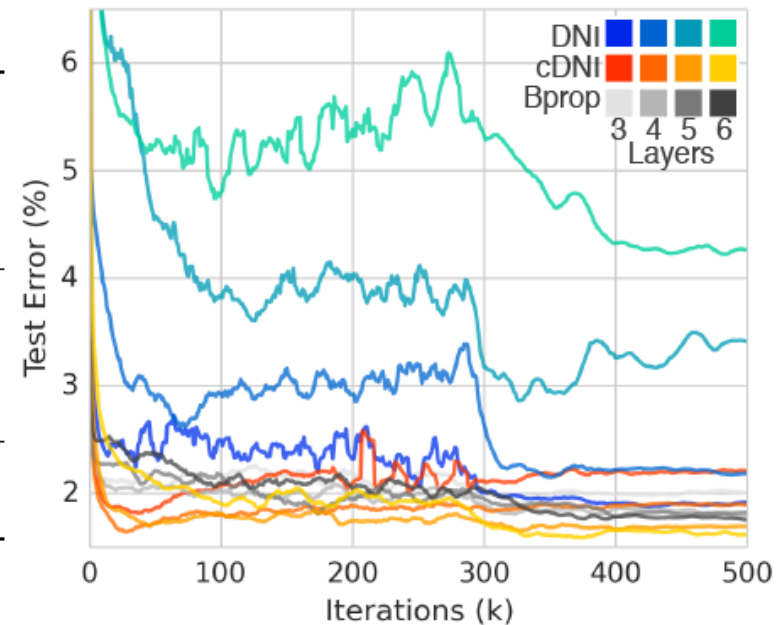
- Avantages du DNI:
- Désynchronisation de la backpropagation
 - Adaptation des updates de gradient à l'importance des niveaux
 - Une étape obligée vers l'apprentissage continu
 - Un parallèle biologique crédible pour la backpropagation

DNI – Exemple sur MNIST et CIFAR



Layers		MNIST (% Error)				CIFAR-10 (% Error)			
		No Bprop	Bprop	DNI	cDNI	No Bprop	Bprop	DNI	cDNI
FCN	3	9.3	2.0	1.9	2.2	54.9	43.5	42.5	48.5
	4	12.6	1.8	2.2	1.9	57.2	43.0	45.0	45.1
	5	16.2	1.8	3.4	1.7	59.6	41.7	46.9	43.5
	6	21.4	1.8	4.3	1.6	61.9	42.0	49.7	46.8
CNN	3	0.9	0.8	0.9	1.0	28.7	17.9	19.5	19.0
	4	2.8	0.6	0.7	0.8	38.1	15.7	19.5	16.4

Gagnants

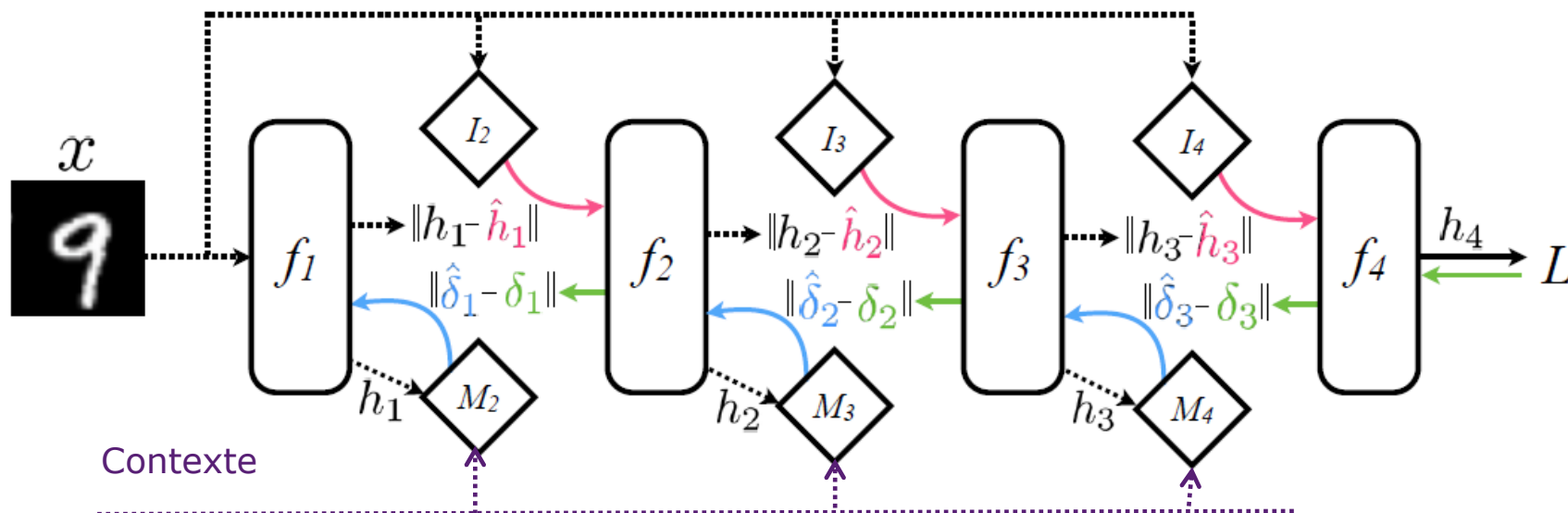


cDNI : Gradient synthétique conditionnel

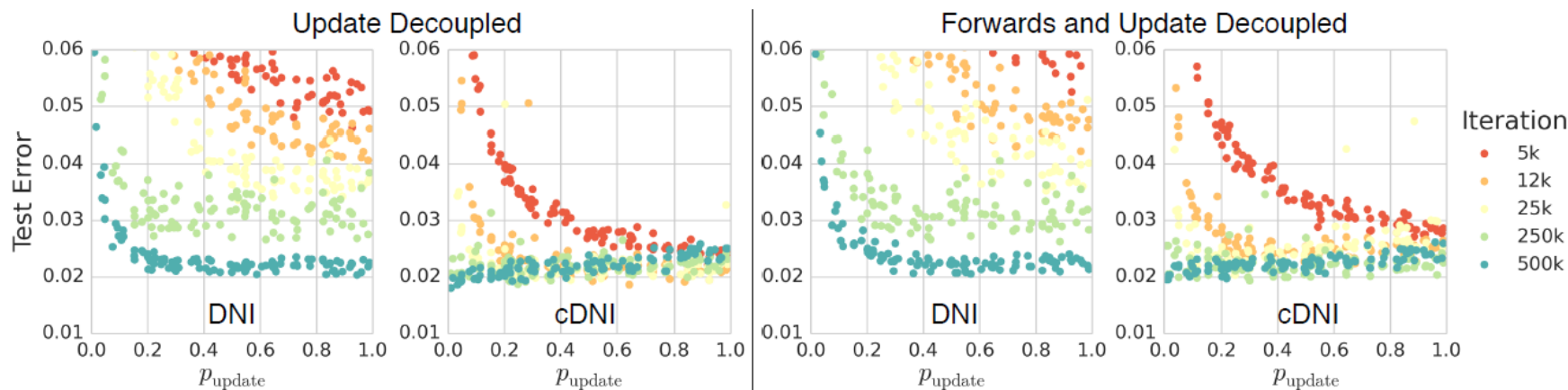
Leçon : Le gradient doit être considéré comme un nœud à part entière

Avantage du cDNI: Il augmente la précision du résultat en régularisant la convergence sans biais

DNI généralisé



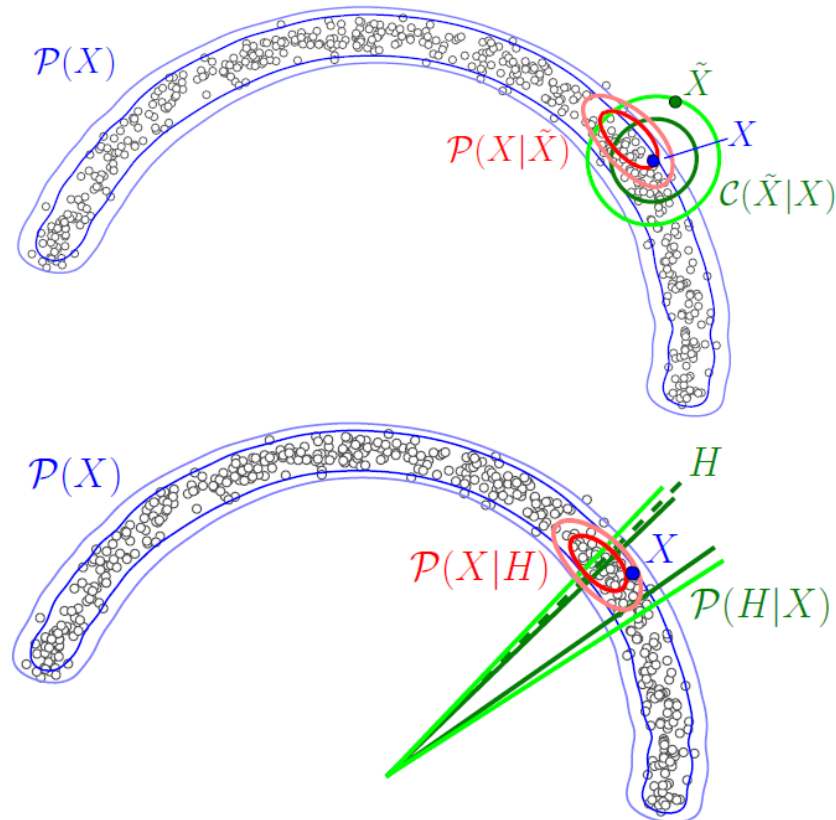
Tous les gradients et le forwards sont nodisés et conditionnels



L'hypothèse de variété non supervisée

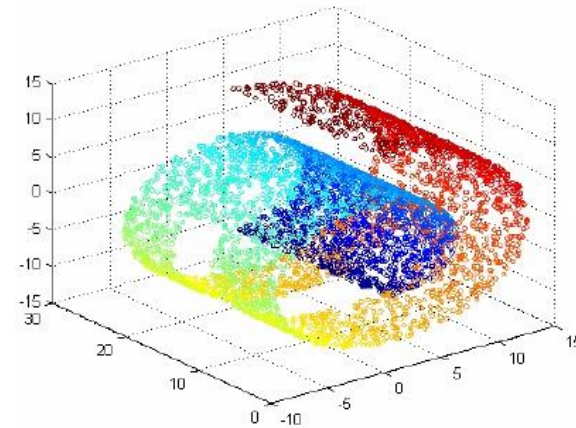
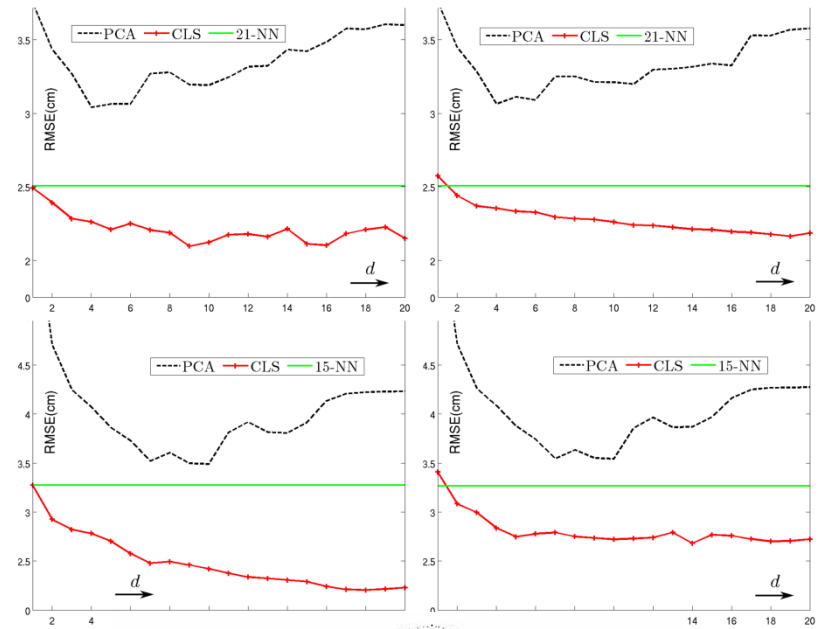
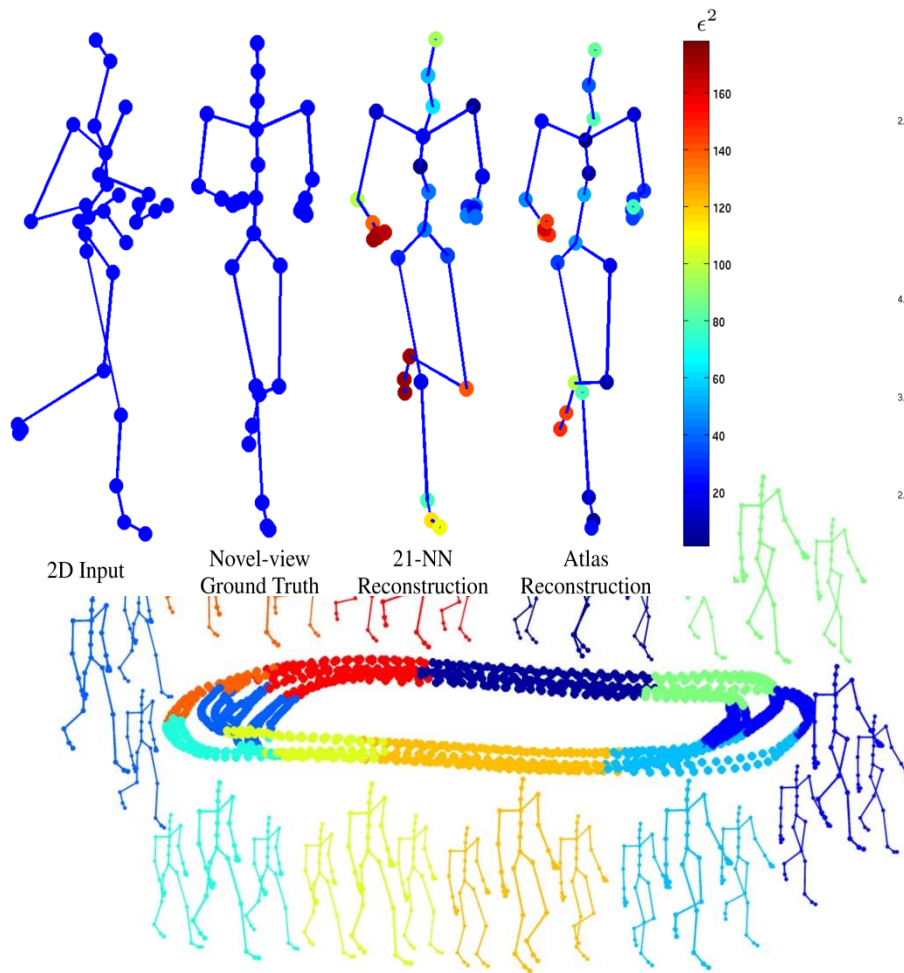


Hypothèse : Les données, quoique ensemble de points dans un espace de grande dimension, sont groupée selon une sous variété de faible dimension intrinsèque

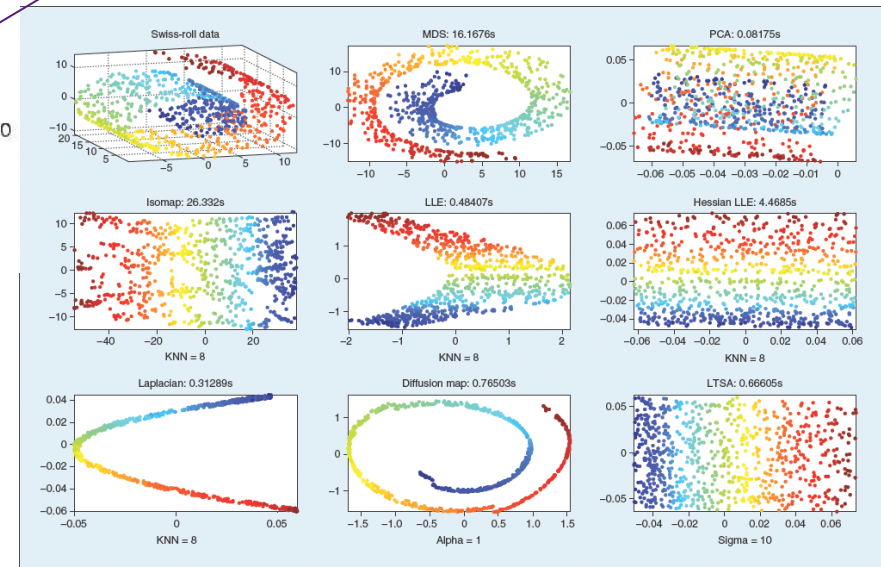
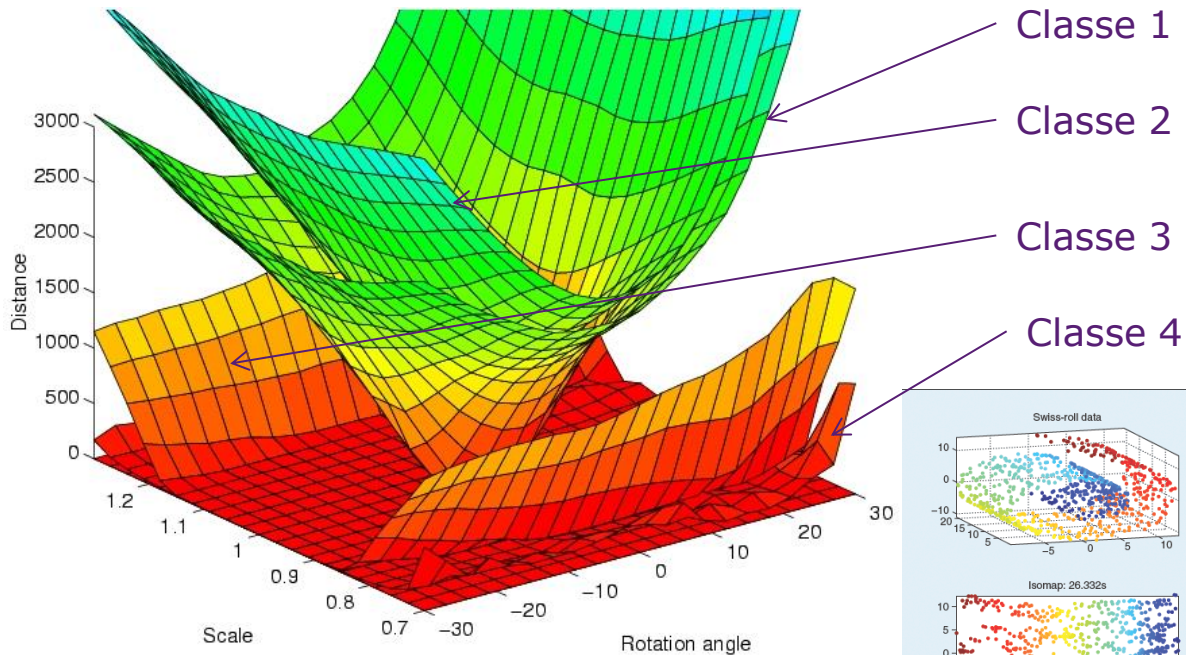


<http://colah.github.io/posts/2014-03-NN-Manifolds-Topology/>

Exemple de sous-variété



Le classifieur de sous-varieté tangente



Minimisation de l'autoencodeur associé



$$r = \text{Enc} \circ \text{Dec} \qquad J(x) = \frac{\partial \text{enc}(x)}{\partial x}$$

$SVD_i[M]$ est le i ème plus important vecteur de la décomposition singulière de M

1^{ier} ordre Contrastive $\mathcal{L}_{CAE} = (\|r(x) - x\|^2 + \lambda \|J(x)\|^2)$

2^{ieme} ordre stochastique Contrastive

$$\mathcal{L}_{CAE} = \left(\|r(x) - x\|^2 + \lambda \|J(x)\|^2 + \gamma \mathbb{E}_{\epsilon \sim \mathcal{N}(0, \sigma^2 I)} [\|J(x) - J(x + \epsilon)\|] \right)$$

Emule un hessien

1^{ier} ordre stochastique MTC

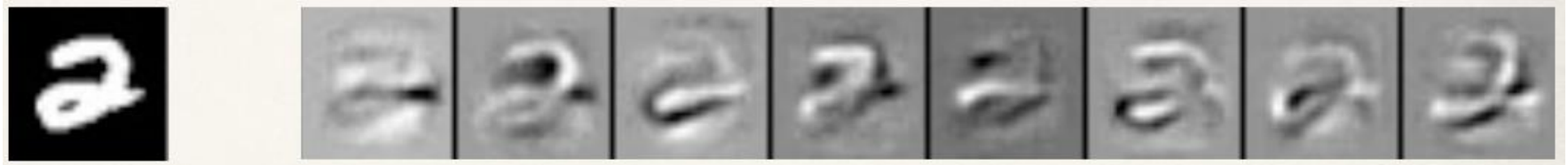
$$\mathcal{L}_{CAE} = \left(\|r(x) - x\|^2 + \lambda \left\| \sum_{i=1}^n SVD_i[J(x)] \right\|^2 \right)$$

MTC sur MNIST



input

Les différents vecteurs tangents



MTC sur CIFAR10



input

Les différents vecteurs tangents



↑
PCA



↑
Autoencodeur contractif

MTC: performances



K-NN	NN	SVM	DBN	CAE	DBM	CNN	MTC
3.09%	1.60%	1.40%	1.17%	1.04%	0.95%	0.95%	0.81%

MNIST : Modèles invariants par permutation

Prévision financière

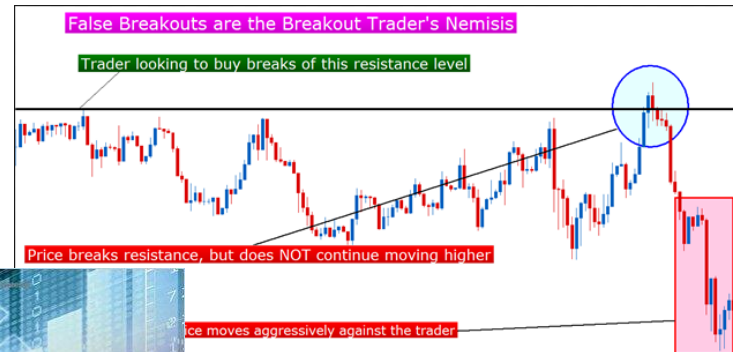
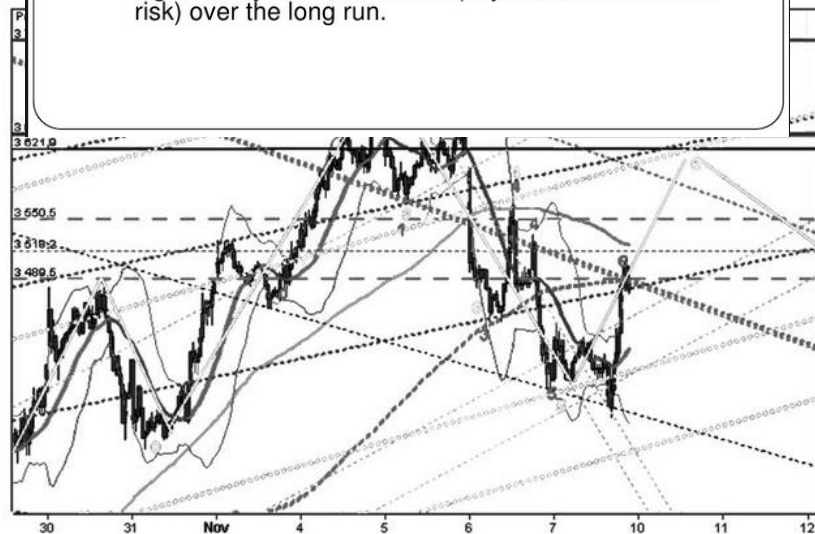


Prévision : l'Age de la magie

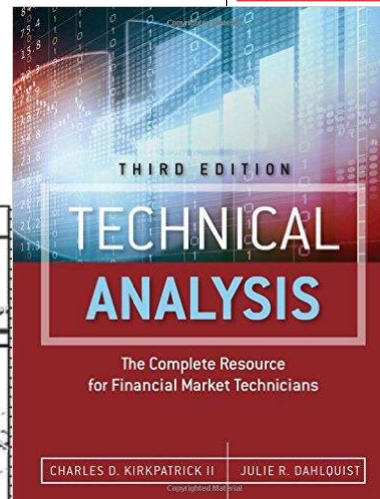


Summary of Tests of the EMH

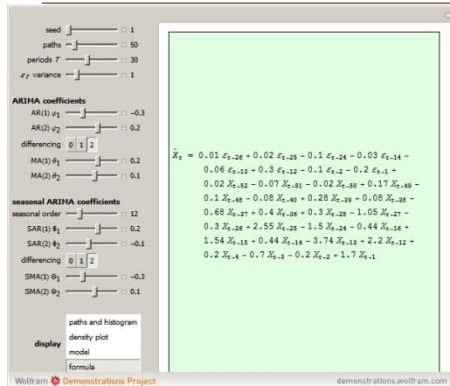
- Weak form is supported, so technical analysis cannot consistently outperform the market.
- Semi-strong form is mostly supported, so fundamental analysis cannot consistently outperform the market.
- Strong form is generally not supported. If you have secret ("insider") information, you CAN use it to earn excess returns on a consistent basis.
- Ultimately, most believe that the market is very efficient, though not perfectly efficient. It is unlikely that any system of analysis could consistently and significantly beat the market (adjusted for costs and risk) over the long run.



Price moves aggressively against the trader

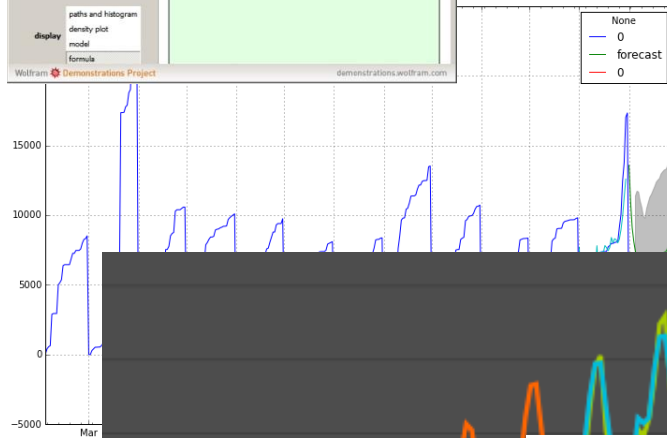


Prévision : Vision mathématique pure



$$(1 - \phi_1 B) (1 - \Phi_1 B^4) (1 - B) (1 - B^4) y_t = (1 + \theta_1 B) (1 + \Theta_1 B^4) e_t.$$

(Non-seasonal AR(1))
(Non-seasonal difference)
(Non-seasonal MA(1))
(Seasonal MA(1))



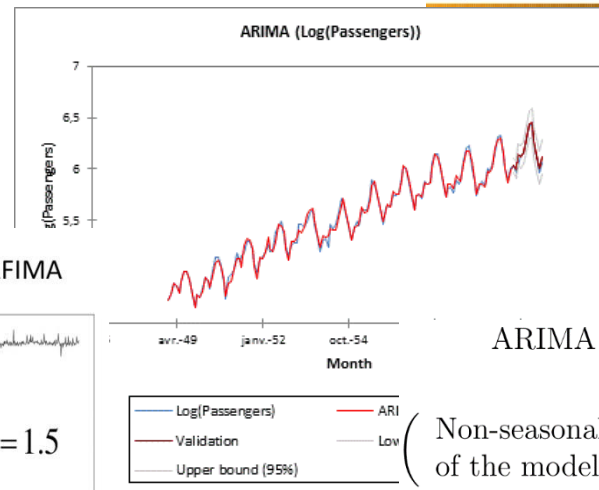
ARMA(p,q) Models

- A model with autoregressive terms can be combined with a model having moving average terms to get an ARMA(p,q) model:

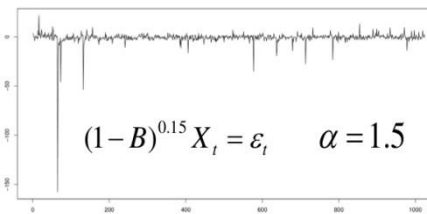
$$Y_t = \phi_0 + \phi_1 Y_{t-1} + \dots + \phi_p Y_{t-p} + \varepsilon_t - \theta_1 \varepsilon_{t-1} - \theta_2 \varepsilon_{t-2} - \dots - \theta_q \varepsilon_{t-q}$$
- ARMA(p,q) models can describe a wide variety of behaviors for stationary time series.
- Theoretical behavior of the ACF and PACF for autoregressive-moving average processes:

	ACF	PACF
AR(p)	Die out	Cut off after the order p of the process
MA(q)	Cut off after the order q of the process	Die out
ARMA(p,q)	Die out	Die out

Note that:
- ARMA(p,0) = AR(p)
- ARMA(0,q) = MA(q)
In practice, the values of p and q each rarely exceed 2.
In this context...
- "Die out" means "tend to zero gradually"
- "Cut off" means "disappear" or "is zero"



Pure symmetric α -stable ARFIMA



ARIMA $(p, d, q) (P, D, Q)_m$

(Non-seasonal part of the model)
(Seasonal part of the model)

Prévision : martingalité orthodoxe



TESTING THE MARTINGALE HYPOTHESIS*

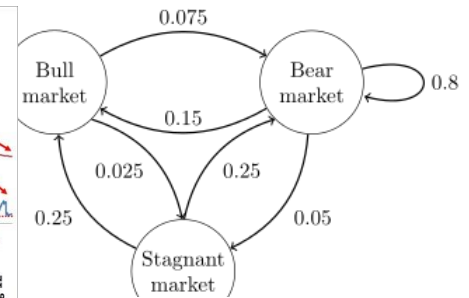
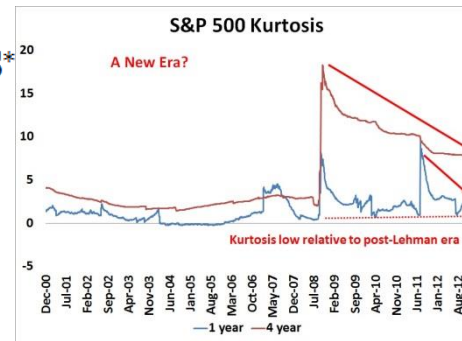
J. CARLOS ESCANCIANO

Indiana University, Bloomington, USA

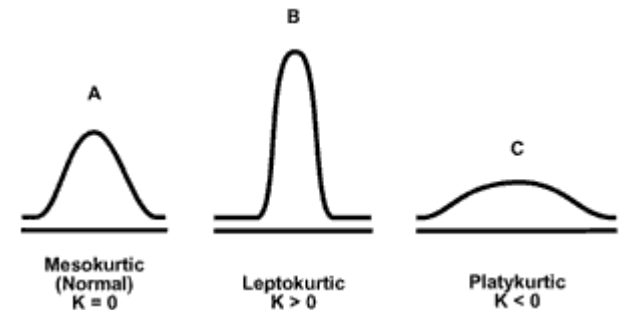
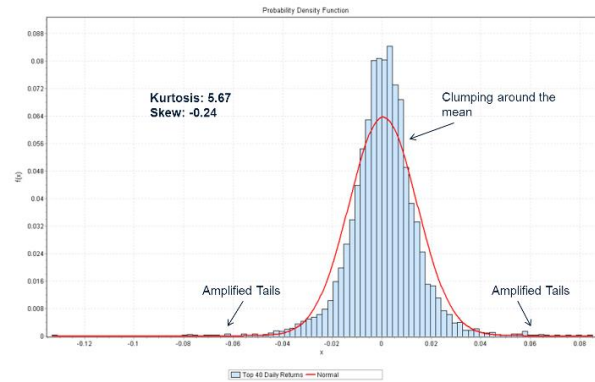
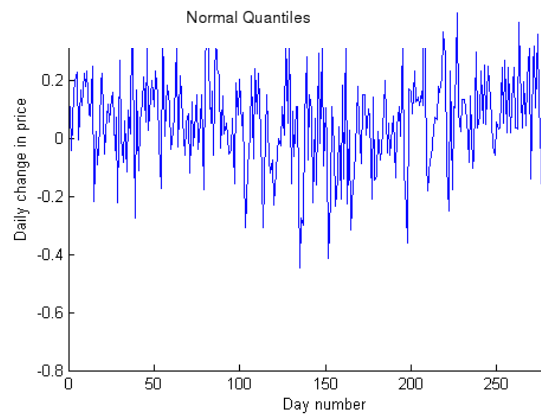
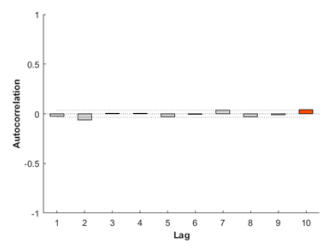
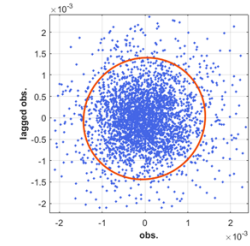
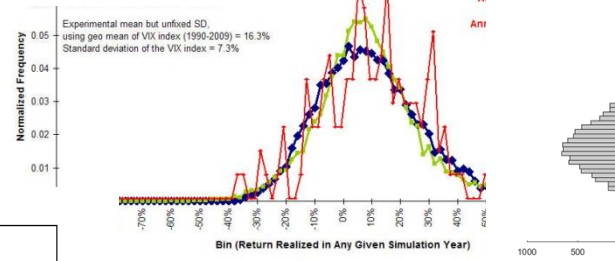
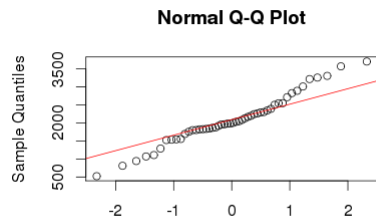
IGNACIO N. LOBATO

Instituto Tecnológico Autónomo de México, México D.F., Mexico

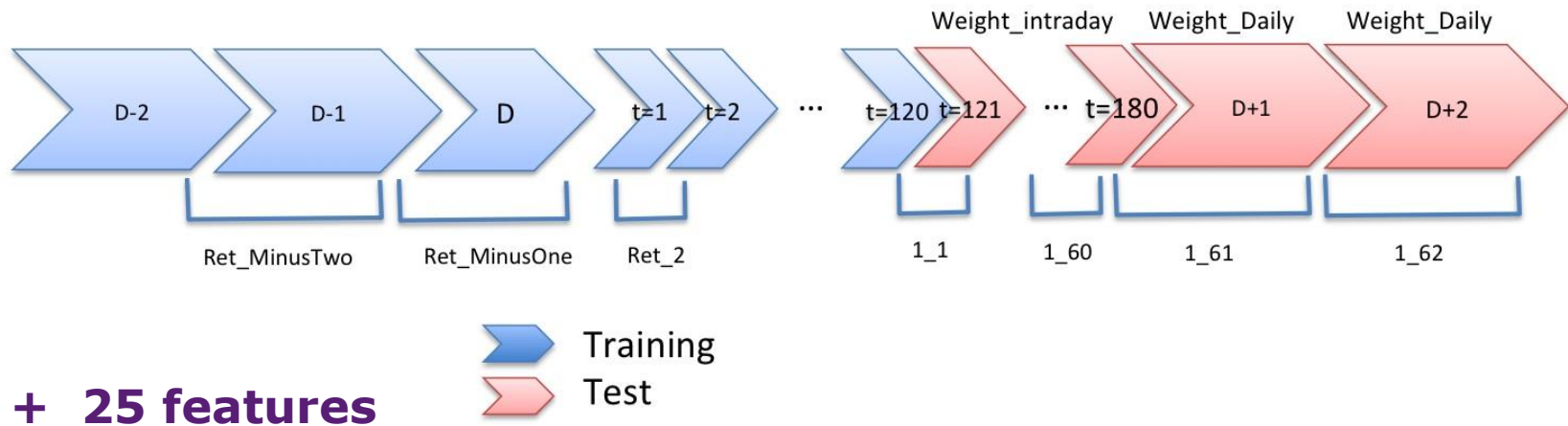
October 25, 2007



Invariance test on the daily changes in the 10-year yield to maturity



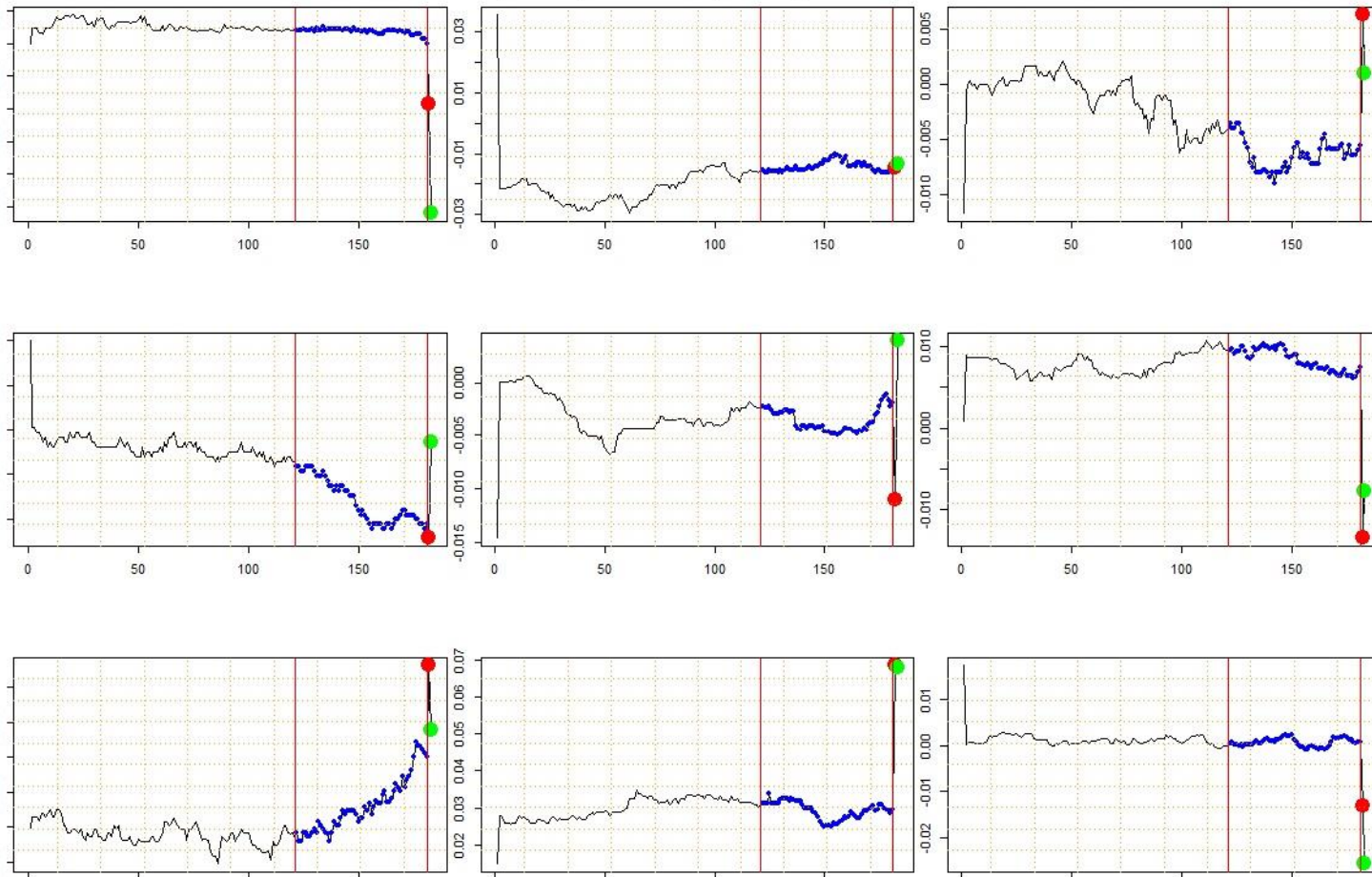
Winton Stock Market Challenge



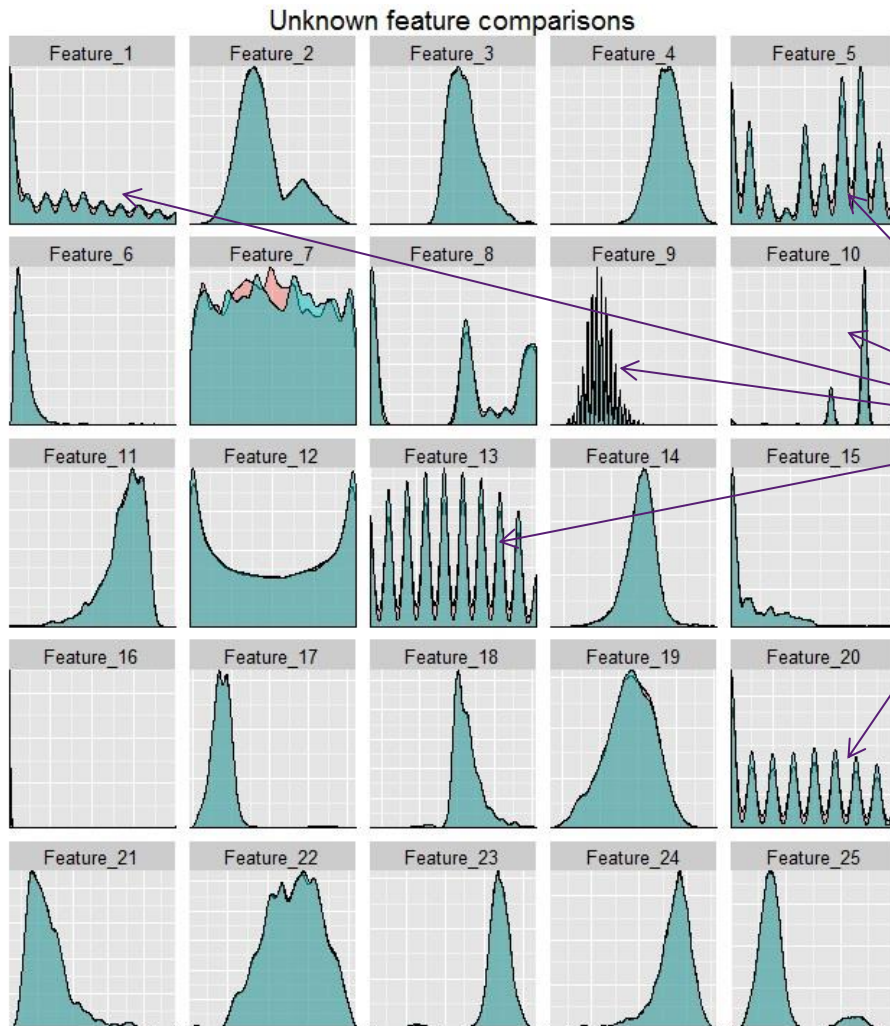
Training : 40000 séries

Notation : 100000 séries

Winton Stock Market Challenge (2)

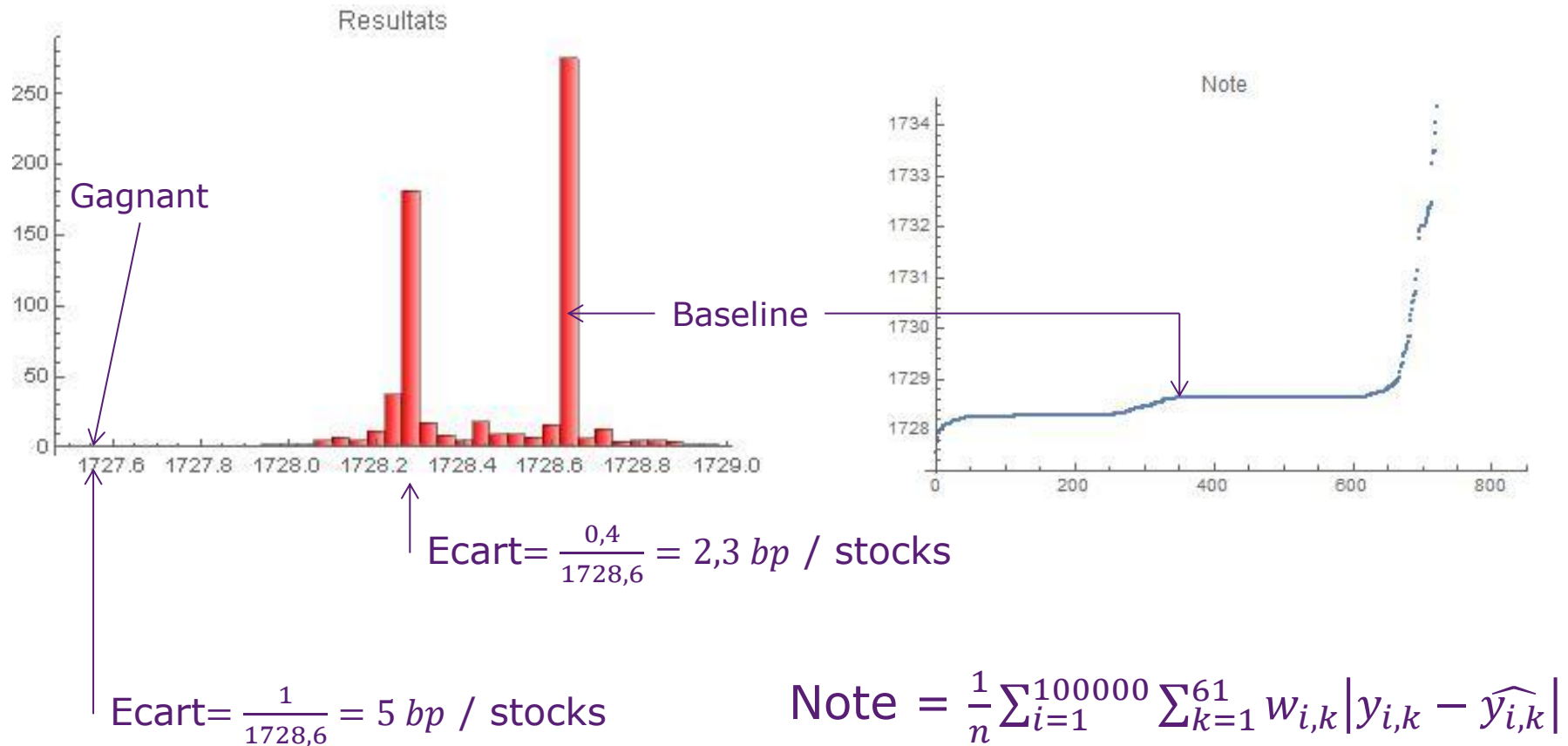


Winton Stock Market Challenge (3)



Certaines features
sont évidemment catégorielles

Winton Stock Market Challenge: résultats



Stock Market Challenge : ma solution



Une série boursière est comme un discours dont il faut apprendre la langue

```
model:add(SliceList)
if dropoutFlag then model:add(nn.Sequencer(nn.Dropout(dropoutProb))) end
model:add(nn.Sequencer(nn.FastLSTM(embeddingSize+nbfeatures, embeddingSize, rhoOutput)))
model:add(nn.Sequencer(nn.Linear(embeddingSize, dictionarySize))
if not(CrossEntropyFlag) then model:add(nn.Sequencer(nn.LogSoftMax())) end
return model
```

```
taille du dataset de training :36
error goal 0.2 and max iteration = 20000
error for iteration 500 is 1.8108169855281
error for iteration 1000 is 1.5390473429868
error for iteration 1500 is 1.3987408592645
error for iteration 2000 is 1.2643873510552
error for iteration 2500 is 1.190822624829
error for iteration 3000 is 1.0666057409108
error for iteration 3500 is 0.92792516314329
error for iteration 4000 is 0.80726509513093
error for iteration 4500 is 0.76559975886313
error for iteration 5000 is 0.77901771165743
error for iteration 5500 is 0.66204038208886
error for iteration 6000 is 0.57159945965616
error for iteration 6500 is 0.64896469790744
error for iteration 7000 is 0.76506822398161
```

```
Module[{j0 = 1, patternSet},
  patternSet = Union[
    Table[With[{j1 = 7, k = k0}, Function[{set, item},
      (set[[item, 1]] = j1) && set[[item, 13]] = k]], {k0, 1, 3}];
  BuildAndWriteReducedSets[patternSet]
]
_1_ train,test written ={36, 133}
_2_ train,test written ={36, 124}
_3_ train,test written ={44, 122}
```

```
error for iteration 11500 is 0.33068015678106
error for iteration 12000 is 0.28179228332878
error for iteration 12500 is 0.23297878925448
global error = 0.19949678173875
time for learning :5504.9666700363 sec
```

LSTM

$$1728.62346 - 1728.62301 = \frac{100\,000}{36}$$

1.25

$$1728.62301 - \frac{1728.62346 - 1728.62301}{36} \cdot 100\,000$$

1727.37

Mon score pour 36 stocks

Mon score théorique pour les 100000 stocks

1 — * Just Pay your Bills 1727.53576

2 — * Humberto Brandão 1727.73786



Zero prediction

1728.62346



Prédire des cours de bourse avec des news



Table 2: Weekly Summary Statistics by Market Capitalization

Decile	Log market cap	News stories per week	Proportion of firms without news	Return	Return w/news	Return w/o news	Difference	t-statistics
Smallest 1	9.42	0.16	0.95	-0.14%	2.00%	-0.24%	2.24%	3.47
2	10.56	0.59	0.94	-0.12%	1.51%	-0.23%	1.75%	3.59
3	11.22	1.33	0.92	-0.14%	-0.02%	-0.15%	0.12%	0.40
4	11.80	2.49	0.89	-0.06%	-0.04%	-0.06%	0.01%	0.04
5	12.35	4.01	0.87	-0.05%	-0.28%	-0.01%	-0.26%	-0.86
6	12.88	5.81	0.85	-0.01%	-0.26%	0.03%	-0.29%	-0.99
7	13.43	7.63	0.82	0.04%	-0.07%	0.06%	-0.13%	-0.48
8	14.04	10.19	0.76	0.05%	0.08%	0.04%	0.04%	0.15
9	14.83	13.30	0.66	0.09%	0.12%	0.07%	0.05%	0.25
Largest 10	16.48	22.42	0.34	0.06%	0.07%	0.04%	0.04%	0.18

Notes: This table presents weekly statistics for decile portfolios grouped on market capitalization over calendar years 2003 to 2010 (417 weeks). We divide the firms into deciles based on the market capitalization at the beginning of the month. Each week we note the number of news articles with relevance of at least 0.35, proportion of firms without news, average return, average return for firms with news, average return for firms without news, difference of return between firms with and without news and the *t*-statistics for the difference.

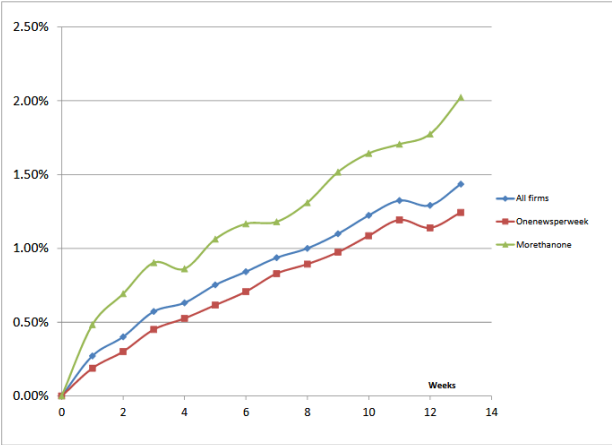


Figure 4: Cumulative Weekly Post-News Long-Short Quintile Returns

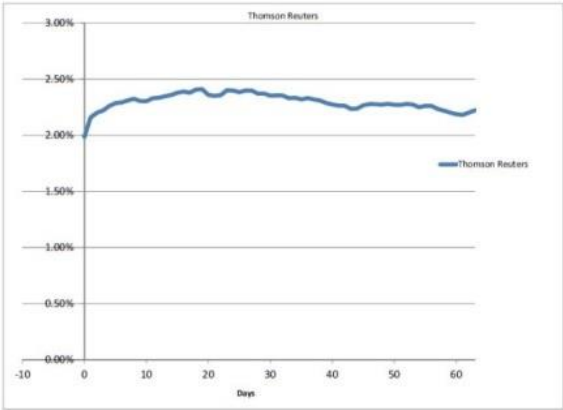


Figure 1: Cumulative Daily News and Post-News Long-Short Quintile Returns

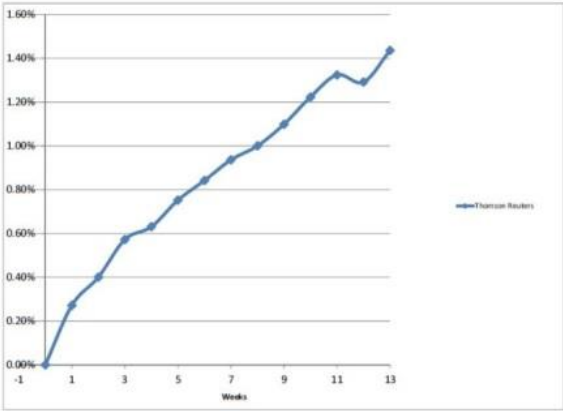
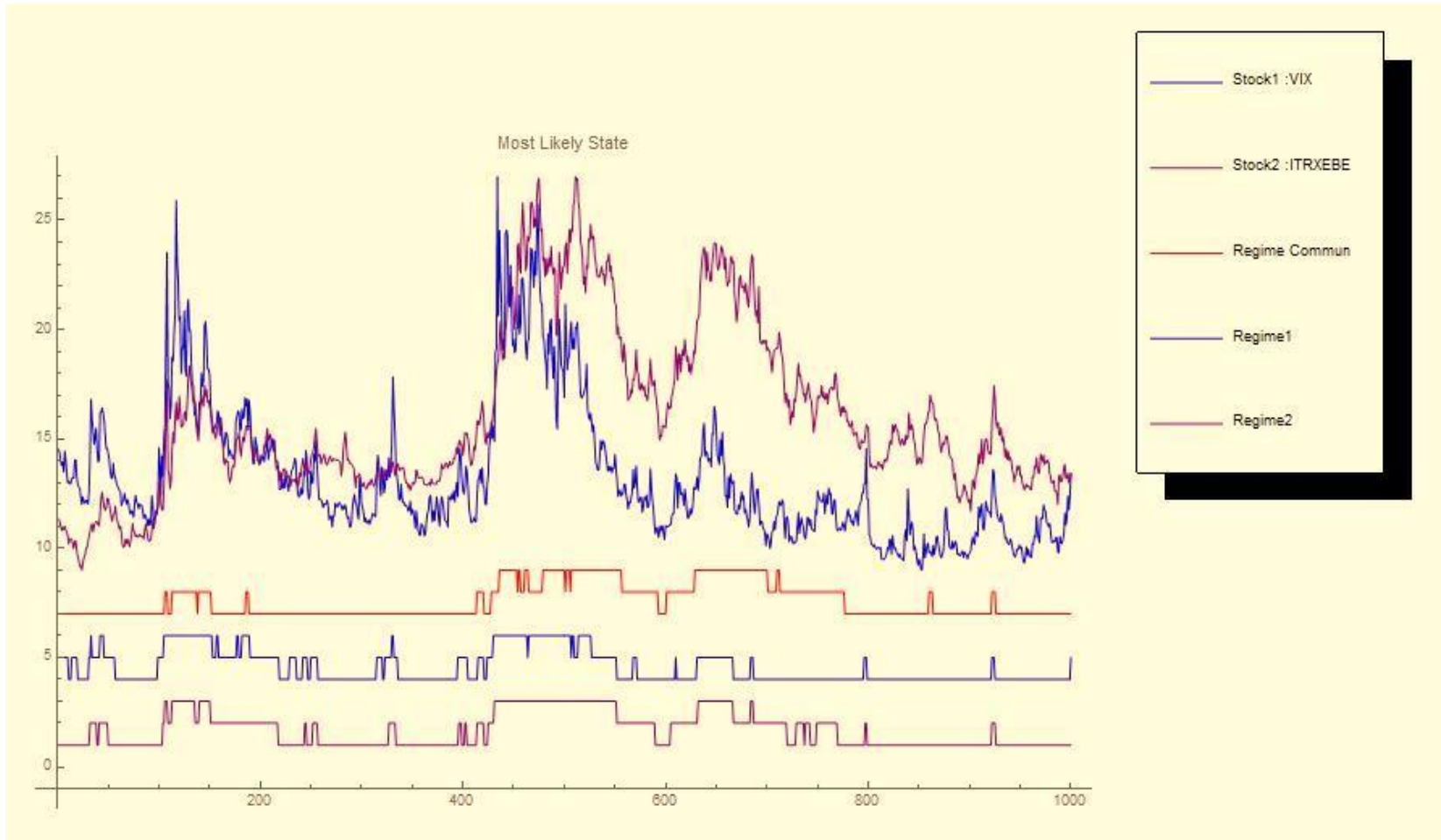
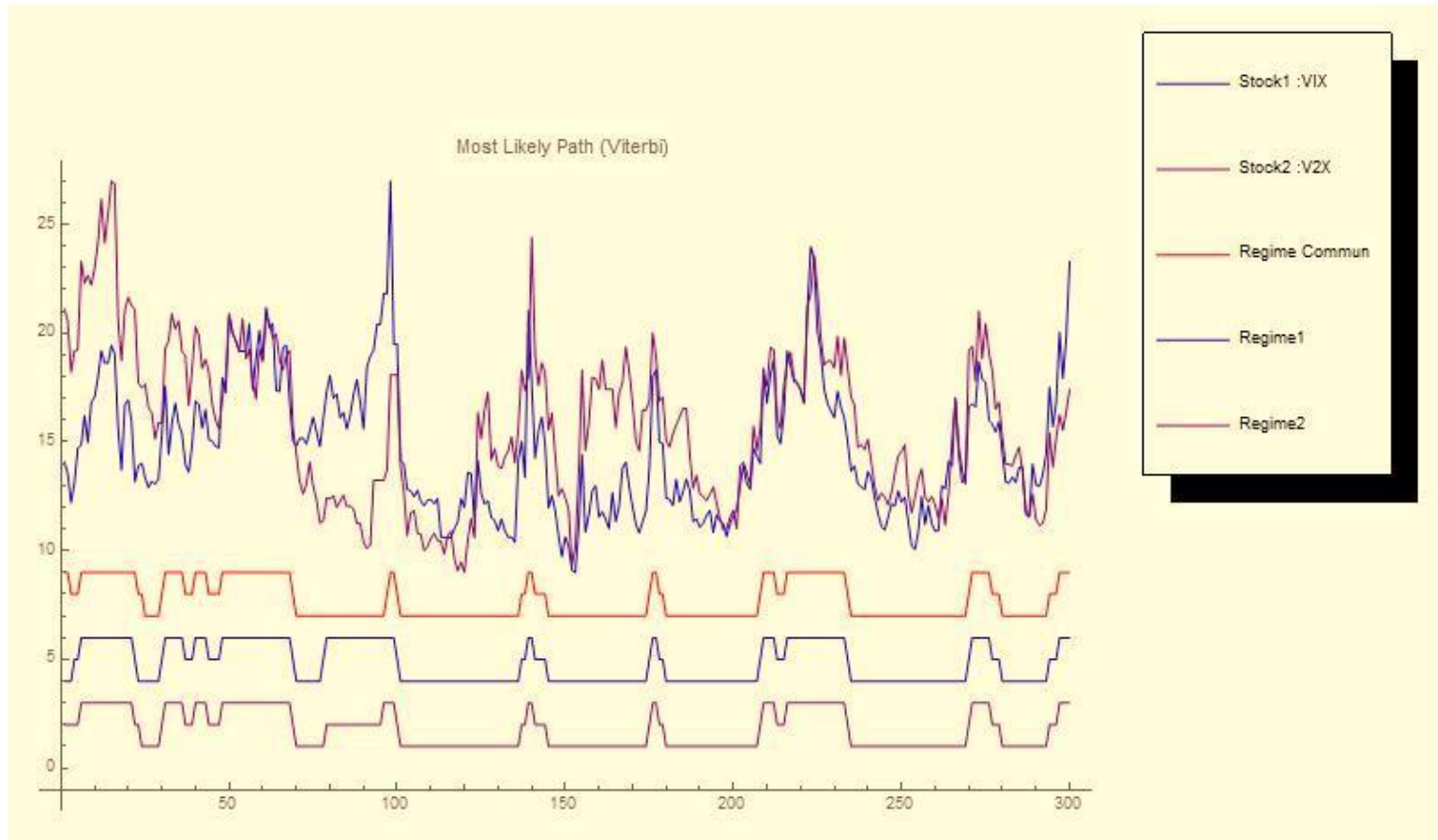


Figure 2: Cumulative Weekly Post-News Long-Short Decile Returns

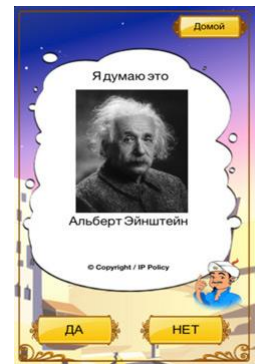
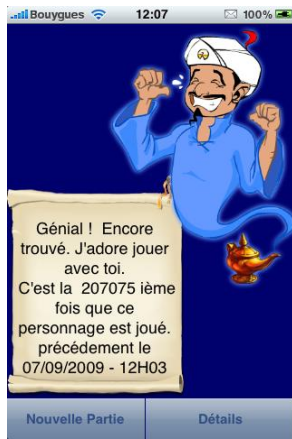
HMM multi-dimensionnel (VIX/ITERXEBE)



HMM multi-dimensionnel (VIX/V2X)



Akinator : un succes planétaire



Plus de 100000 personnages !
14 langues !

Akinator : le principe



Bayes :

$$P(S = S_k | G = G_n) = \frac{P(G = G_n | S = S_k) P(S = S_k)}{P(G = G_n)}$$

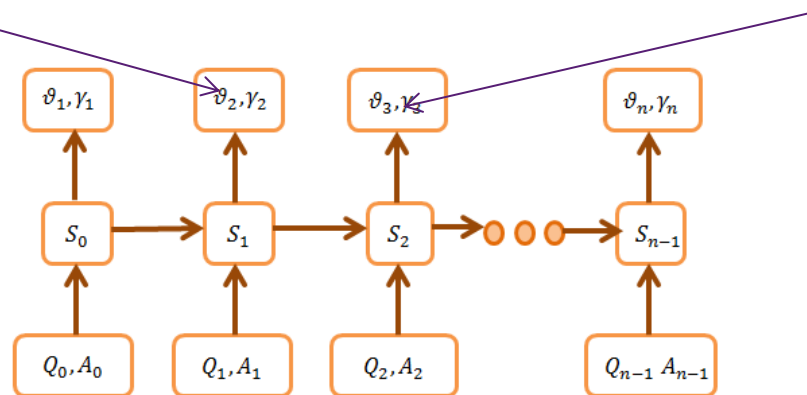
Maximiser l'entropie croisée

$$H(d_{before}, d_{after}^{q_n}) = \int_S (d_{before}(S) \log [d_{after}^{q_n}(S)]) dS$$

$$\{P(g_n = \{q, a\} | G_i)\}_{r \in R}$$

$$\{P(g_i | S = S_k, G = G_{i-1})\}_{k \in K}$$

LSTM :





Interpolateur / extrapolateur pour série financière à réalité augmentée

- **Tirant parti de l'existence**
 - d'une grande quantité de séries synchrones
 - de séries de données non numériques
- Capable de tirer parti de l'existence de corrélations complexes (spatiales et temporelles) non simplement exprimables par des concepts mathématique usuels
- Tirant parti de la mise a disposition de grandes puissances de calculs (GPU, clusters ..)
- Utilisant les concepts de réseau de neurones précédemment exposés

Cœur de l'architecture du réseau idéal



Choix architecturaux

Génératif

Réactivité a la complexité

Mémoire Differentiable

Grande sensibilité au passé

Manifold Tang. Cl.

Profondeur de complexité explorée

Interface Neur. Déc.

Modulariser et distribuer l'apprentissage du réseau



- **Architecture du réseau neuronal est un facteur clé**
 1. Réseau profond
 2. Génératif / discriminatif
 3. Inférence variationnelle
 4. Mécanisme d'attention
 5. Mémoire différentiable
 6. Découplage des interfaces neuronaux
 7. Contrôle des sous-variétés du gradient
- **Les performances restituées par réseau neuronal correctement pensé sont souvent suprahumaines**
- **La prévision de séries financières en réalité augmentée est possible, a condition qu'elle soit**
 1. Probabiliste
 2. A input large.

Thank you for your time

